Parametric Part Modeling Field Guide

A practical workflow reference for creating editable mechanical 3D parts in SOLIDWORKS. Prepared as searchable test data for the Atlas AI retrieval-augmented generation (RAG) module.

Document purpose: Teach a repeatable feature-based modelling process: plan the part, create stable sketches, build features, inspect the model, and prepare manufacturing outputs.

Document control

Field	Value
Document ID	ATLAS-SW-PART-001
Version	1.0
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Primary software context	SOLIDWORKS desktop, 2025-era interface; feature names are broadly applicable across recent releases
Units used in examples	Millimetres (mm), degrees, grams (g), and cubic millimetres (mm^3) unless stated otherwise
Audience	Engineering students, CAD learners, and an AI assistant answering part-modelling questions

Important scope note

This is an original training guide, not an official SOLIDWORKS manual and not a substitute for the help system, company CAD standards, or engineering review. Menu locations and available commands can vary with release, license, workspace, and user interface customisation.

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1. A reliable modelling workflow

A SOLIDWORKS part is normally a history-based parametric model. The FeatureManager design tree records sketches, reference geometry, and features. A good model is not merely a shape that looks correct; it is a controlled set of relationships that can be edited without unexpected failures.

Recommended sequence

- Interpret the design brief. Identify function, interfaces, loads, moving clearances, manufacturing process, material, and required deliverables.
- Choose a coordinate system. Put the origin at a functional datum such as a symmetry centre, mounting face, shaft axis, or primary interface.
- Select the first feature. Use a simple base feature that captures the dominant form and gives later features stable references.
- Build sketches with explicit dimensions and geometric relations. Fully define critical sketches where practical.
- Add features in functional order: primary volume, major cuts, holes, ribs or webs, then fillets and chamfers.
- Inspect the FeatureManager tree and rebuild. Resolve warnings before adding more dependent features.
- Check dimensions, mass properties, interference where relevant, and manufacturability.
- Create drawings and export neutral files only after the native model is stable.

> Design rule: Model from functional datums and interfaces. Avoid choosing edges or faces merely because they are convenient if those references are likely to disappear when upstream geometry changes.

Feature tree strategy

Use meaningful feature and sketch names, for example BasePlate, MountingHoleLayout, ClearanceCut, Rib_Web, and EdgeBreak. Names should describe purpose, not only command type. Keep related features near one another and avoid a long sequence of opaque Sketch17 / Cut-Extrude9 names.

Stage	Typical operation	Reason
Primary shape	Extruded Boss/Base, Revolved Boss/Base	Creates the main solid volume
Major removal	Extruded Cut, Revolved Cut	Defines pockets, slots, bores, and clearances
Functional details	Hole Wizard, ribs, patterns	Adds interfaces consistently
Edge treatment	Fillet, Chamfer	Improves safety, assembly, and finishing
Validation	Measure, Mass Properties, section view	Checks geometry and design intent

2. Sketches and design intent

A sketch is a 2D profile or layout used to define a feature. Sketch quality strongly affects model editability. A profile can be geometrically closed yet still be poorly constrained, or fully constrained yet dimensioned in a way that does not reflect design intent.

Sketch entities and relations

- Common entities: line, centreline, rectangle, circle, arc, slot, spline, polygon, and point.
- Common relations: coincident, horizontal, vertical, parallel, perpendicular, tangent, concentric, equal, midpoint, collinear, and symmetric.
- Use construction geometry for centre axes, symmetry lines, pitch circles, and layout references. Construction geometry normally does not create the solid profile boundary.
- Use dimensions to communicate size and position; use relations to communicate geometric behaviour. Do not replace every relation with a separate dimension.

- Avoid redundant dimensions. A conflicting or over-defined sketch usually indicates that a relation or dimension duplicates an existing constraint.

Under-defined, fully defined, and over-defined sketches

State	Meaning	Recommended response
Under-defined	One or more degrees of freedom remain; geometry may move.	Add the missing dimensions or relations where the geometry is functionally important.
Fully defined	No remaining degrees of freedom in the active sketch.	Preferred for critical production geometry, provided the dimensions reflect design intent.
Over-defined	Conflicting or redundant constraints exist.	Inspect recently added dimensions and relations; remove duplication rather than forcing a value.

Practical sketch workflow

- Select the intended plane or a stable planar face, then start a sketch.
- Add the simplest profile first. Use centreline symmetry when the part is symmetric about a functional axis.
- Apply geometric relations before adding all dimensions.
- Dimension from datums or functional references. Avoid chains of dimensions that accumulate tolerance unless required by the design.
- Use SketchXpert or sketch diagnostics when relations conflict; inspect the actual constraint logic before accepting automatic changes.
- Exit the sketch and preview the feature before committing it.

Example: For a 100 mm by 60 mm rectangular plate centred on the origin, use a centre rectangle and dimension its overall width and height. The origin remains the design centre, so later symmetric features can reference it.

3. Core solid features

Core features turn sketches or reference curves into solid geometry. Select the feature based on the geometry and manufacturing logic, not simply on which command is fastest.

Feature	Input	Typical use	Key controls
Extruded Boss/Base	Closed 2D sketch	Plates, blocks, pads, webs	End condition, depth, direction, draft, merge result
Extruded Cut	Closed 2D sketch	Pockets, slots, profiles, windows	Blind, Through All, Up to Surface/Next, direction
Revolved Boss/Base	Profile plus axis	Shafts, hubs, pulleys, turned forms	Axis, angle, direction, thin feature
Revolved Cut	Profile plus axis	Grooves, turned recesses, internal profiles	Axis and revolve angle
Sweep	Profile and path	Pipes, handles, curved sections	Path, orientation, twist control
Loft	Two or more profiles	Transition shapes, ducts, tapered forms	Profile order, guide curves, start/end constraints

Extruded Boss/Base procedure

- Create a closed sketch on a suitable plane or planar face.
- Choose Features → Extruded Boss/Base (or the equivalent command in the current interface).
- Set the end condition. Blind uses a numeric depth; Mid Plane extrudes equally on both sides; Through All extends through the available model; Up to Surface stops at a selected face or surface.
- Check direction, merge result, and preview. If the result should be a separate body, confirm that merging is disabled where appropriate.

- Accept, rename the feature, and verify the resulting body.

Extruded Cut procedure

- Create a closed sketch that intersects the target solid.
- Start Extruded Cut and choose the direction and end condition.
- Use Through All for a true through-hole or through-slot when the design intent is to pass completely through the body.
- Use Blind for a controlled depth; document the depth as a functional requirement.
- Preview from multiple views to ensure the cut removes the intended material and does not pass through an unintended wall.

Revolve, sweep, and loft guidance

For a revolve, the profile must be positioned relative to an axis of revolution. Check that the profile does not create invalid self-intersecting or zero-thickness geometry. For a sweep, use a clean path and a profile positioned at the path start; inspect section orientation on tight bends. For a loft, keep profile order consistent and use guide curves only when they control a genuine shape requirement. Poorly aligned connectors can twist or distort the loft.

> Feature selection shortcut: Use Extrude for prismatic forms, Revolve for rotationally symmetric forms, Sweep for a constant or controlled section following a path, and Loft for transitions between different sections.

4. Secondary and finishing features

Secondary features add local geometry after the primary shape is stable. Their order matters because features depend on upstream faces, edges, sketches, and bodies.

Feature	Purpose	Modelling guidance
Fillet	Rounds selected edges or faces	Add late where practical. Use a radius tied to function, strength, comfort, or manufacturing capability.
Chamfer	Bevels an edge	Use equal-distance, distance-angle, or face-face definitions as appropriate. Often used for deburring and lead-ins.
Shell	Creates a hollow part with specified wall thickness	Select faces to remove and inspect all wall transitions and internal corners.
Rib	Adds thin reinforcing material	Sketch the rib centreline/profile and define thickness and direction carefully. Avoid abrupt load paths.
Draft	Adds taper to faces	Use a neutral plane and draft direction consistent with tooling or release direction.
Linear/Circular Pattern	Repeats features or bodies	Pattern the feature that best represents design intent; verify spacing, count, and direction.
Mirror	Reflects geometry about a plane	Use a stable centre plane or datum plane; decide whether to mirror features or bodies.

Why fillets are often added late

Fillets can change or split edge topology. If an early fillet is referenced by later sketches or features, a small upstream modification may cause those references to fail. A robust sequence is to create the main geometry, functional cuts, and patterns first, then apply cosmetic or edge-break fillets near the end. This is a guideline, not an absolute rule: a fillet that defines the functional shape may need to be created earlier.

Shelling checklist

- Confirm the target body and identify the opening face or faces.

- Choose wall thickness based on load, material, manufacturing process, and part size; do not copy a generic value blindly.
- Inspect corners, narrow regions, and intersections where offset faces may collide.
- Preview the shell and use section views to check internal wall continuity.
- If the shell fails, simplify corner geometry, reduce thickness, or shell before adding problematic details.

5. Holes, threads, and patterns

Holes are often functional interfaces for fasteners, dowels, shafts, pins, and fluid or cable routing. A hole should be specified by its function, not only by its visible diameter.

Hole Wizard workflow

- Open Hole Wizard and select the hole category and standard supported by the project requirements.
- Choose the hole type: simple, clearance, tapped, counterbore, countersink, or other available type.
- Set size, fit/class, end condition, and any thread or drill-point options relevant to the selected type.
- Open the Positions tab and place each hole centre on the intended face.
- Dimension the hole centres from functional datums or locate them using sketch relations such as symmetry and equal spacing.
- Confirm the preview and inspect the resulting callout in a drawing.

Hole type	Typical purpose	Important distinction
Clearance hole	Allows a fastener shank to pass through	Hole diameter is selected for a fastener clearance condition; it is not the nominal thread diameter.
Tapped hole	Receives a screw with internal threads	Specify thread standard, nominal size, pitch, and usable thread depth.
Counterbore	Recesses a cylindrical fastener head	Specify both pilot-hole size and counterbore diameter/depth.
Countersink	Creates a conical seat	Countersink angle must match the intended screw head standard.
Simple drilled hole	General passage or locating feature	Specify diameter and depth/end condition.

Threads and drawing callouts

A model may represent threads cosmetically rather than with full helical geometry. Cosmetic threads are lighter to rebuild and are usually sufficient for drawings and assembly communication. Full helical threads create actual geometry and may be needed for visualisation, additive manufacture, or special analysis, but they increase file complexity. A tapped-hole note should identify the thread designation and depth; avoid relying on an unlabelled cylinder to communicate a threaded interface.

> Important: Do not assume that a hole with a diameter equal to a screw nominal diameter is a clearance hole. Clearance, tapping drill, and nominal thread diameters are different concepts.

Pattern strategy

Pattern type	Use when	Check
Linear Pattern	Repeated instances along one or two directions	Direction reference, spacing, instance count, skipped instances
Circular Pattern	Features repeated around an axis	Axis selection, total angle, instance count, equal spacing
Sketch-driven Pattern	Instances located by sketch points	Point positions and orientation
Curve-driven Pattern	Instances follow a curve	Curve direction, spacing, alignment and orientation

6. Reference geometry and model robustness

Reference geometry includes planes, axes, points, coordinate systems, and curves. It helps control features independently of incidental model edges and faces.

Prefer stable references

- Use the origin planes and origin axes when they represent the part coordinate system.
- Create named datum planes for mounting interfaces, symmetry planes, offset features, or inspection datums.
- Create axes from cylindrical faces or reference geometry for revolves and circular patterns.
- Use a 3D sketch or reference curve only when the geometry requires it; keep paths continuous and avoid tiny segments.
- Avoid referencing transient edges created by fillets, chamfers, or shell operations unless that edge is explicitly part of the design intent.

Top-down and bottom-up modelling

Approach	Description	Best suited to
Bottom-up	Build each part independently from sketches and dimensions, then assemble parts.	Standalone components with well-defined geometry and interfaces.
Top-down	Create or control components from shared layout geometry, master sketches, skeleton parts, or in-context references.	Mechanisms or products where component sizes and positions are tightly interdependent.

Design intent and dependency control

Design intent expresses how a model should respond to change. For example, four mounting holes should remain symmetric about the centre plane when the plate width changes. Model this with symmetry and shared dimensions rather than four unrelated coordinate values. Use equations or linked values for repeated relationships such as equal wall thickness or a common bolt-circle diameter. Keep external references deliberate and easy to audit.

Common robustness practices

- Use one controlling dimension for a repeated design parameter and link dependent dimensions where appropriate.
- Prefer symmetry, equal, concentric, and midpoint relations to duplicated independent dimensions.
- Create features from functional datums rather than unstable edge selections.
- Rebuild after major operations and inspect the first failed feature in the tree.
- Avoid unnecessary tiny faces, sliver surfaces, zero-thickness conditions, and self-intersections.
- Use configurations for controlled variations such as length, hole spacing, or thickness, not as a substitute for a clear product family strategy.

7. Configurations, drawings, and file exchange

Configurations store alternate states of a part or assembly in one document. They can vary dimensions, feature suppression, properties, and other parameters. Use descriptive configuration names such as SHORT_80, LONG_120, or M6_CLEARANCE rather than Default1.

Drawing preparation

- Create a drawing from the saved native part or assembly.
- Choose a sheet size and projection standard required by the receiving organisation or drawing template.
- Insert views that communicate the shape: front, top, side, isometric, section, detail, and auxiliary views as needed.
- Add dimensions from functional datums. Avoid dimensioning the same feature redundantly unless a specific inspection or reference dimension is required.
- Add hole callouts, thread notes, material, finish, general tolerances, part number, revision, and units.

- Check that the drawing is consistent with the latest model revision and that no dimensions are obscured or ambiguous.

File formats and when to use them

Format	Typical role	What to remember
SLDPRT	Native SOLIDWORKS part	Retains feature tree, sketches, dimensions, and editability.
SLDASM	Native assembly	Retains component relationships and mates; referenced part files must remain available.
SLDDRW	Native drawing	Retains drawing views, dimensions, notes, and sheet information.
STEP (.step/.stp)	Neutral 3D exchange	Usually transfers solid geometry; native feature history is generally not preserved. Confirm AP version and units.
IGES (.igs/.iges)	Surface/curve exchange	Can be useful for surface data but may require healing and is often less robust for solid-part exchange.
STL / 3MF	Mesh-based additive manufacturing	Tessellated output; check units, resolution, watertightness, and orientation.
PDF	Drawing/document sharing	Not an editable 3D model. Use for controlled drawing issue and review.

Export checklist

- Save the native SOLIDWORKS file before exporting.
- Confirm active configuration and body selection.
- Set export units and coordinate orientation intentionally.
- For STEP, choose the required AP option if the receiving system specifies one; verify the exported file by reopening or checking it in a neutral viewer.
- For STL/3MF, set a suitable tessellation tolerance and inspect curved surfaces for visible facets.
- Use meaningful filenames with part number, revision, and format.

8. Troubleshooting failed features

When a feature fails, do not immediately delete it or rebuild the part from scratch. Identify the first feature that fails, read the error message, inspect its input references, and test a controlled change.

Symptom	Likely causes	Diagnostic action / remedy
Extrude fails	Open profile, self-intersection, zero-thickness result, wrong contour selection.	Use sketch diagnostics; check endpoints and overlapping entities; select the intended contour; confirm the feature intersects or merges as expected.
Cut does not remove material	Cut direction points away; profile does not intersect body; wrong end condition.	Reverse direction; inspect preview; use Through All or a suitable end condition.
Fillet fails	Radius too large, short edges, complex corner intersections, invalid edge selection.	Try a smaller radius; split the fillet into separate features; fillet simpler edges first; inspect problematic vertices.
Shell fails	Wall offsets collide; thickness too large; complex corners or small openings.	Reduce thickness; remove or postpone local details; shell before difficult fillets; inspect the selected faces.
Pattern fails	Seed feature is not valid at every instance; direction or axis is wrong; geometry overlaps.	Check preview and pattern direction; use geometry pattern where applicable; skip instances or redesign seed geometry.
Rebuild error after edit	Downstream feature references changed or disappeared.	Find the first red/yellow feature; edit its sketch or reference; replace unstable face/edge references with datum geometry.
Loft or sweep twists	Profile connectors or path orientation are inconsistent.	Align profile start points; simplify guide curves; inspect path continuity; adjust orientation/twist controls.
Imported STEP is not a solid	Gaps, invalid faces, surface body, or incomplete transfer.	Use import diagnostics and healing; knit surfaces where appropriate; confirm the model is closed before treating it as a solid.

A systematic debugging sequence

- Stop at the first failing feature in the FeatureManager tree. Downstream errors may only be consequences.
- Open the feature definition and note its input sketch, face, edge, plane, axis, or body references.
- Suppress later features temporarily if they obscure the problem.
- Edit the sketch or feature with the smallest change that tests the likely cause.
- Rebuild and inspect the whole model, including configurations that use the same feature.
- Rename or document the corrected feature so the reason for the change is clear.

9. Worked example: two-hole mounting bracket

Objective: create a simple L-shaped mounting bracket with a base plate, upright plate, two mounting holes, an optional reinforcing rib, and deburred edges. This example illustrates feature order and modelling intent; the dimensions are illustrative and must be checked against actual loads, fasteners, and manufacturing capability.

Parameter	Illustrative value	Design note
Base plate	100 × 60 × 8 mm	Mounting footprint; centre rectangle about origin.
Upright plate	100 × 50 × 8 mm	Shares the 100 mm width; positioned at one base edge.
Mounting holes	2 × Ø8.5 mm clearance	Illustrative only; confirm fastener and clearance standard.
Hole centres	±30 mm from centreline	Symmetric about the central plane.
Edge chamfer	0.5 mm × 45°	Illustrative deburring feature, not a universal requirement.
Rib	8 mm nominal thickness	Add only if needed after load-path review.

Suggested feature sequence

- Sketch a 100 × 60 mm centred rectangle on the Top Plane; extrude 8 mm to create BasePlate.
- Select a suitable vertical plane or the rear edge face. Sketch the upright profile so it is positioned at the base edge and aligned with the base centreline; extrude 8 mm to create UprightPlate and merge the result.
- Use Hole Wizard to create two Ø8.5 mm clearance holes in the base. Locate their centres symmetrically about the centreline and dimension their offsets from a functional datum.
- If required, create a reinforcing rib between base and upright using a side-plane sketch. Confirm that the rib merges cleanly and does not obstruct fasteners.
- Apply small edge chamfers or fillets after the functional features. Check that the edge treatment does not reduce required contact area.
- Inspect the model in isometric and section views. Use Measure and Mass Properties; create a drawing with overall dimensions and hole callouts.
- Save the editable SLDPRT file and export STEP for neutral exchange. Verify units and reopen the STEP file.

Engineering review required: A bracket is a load-bearing component. The illustrative geometry is not structurally validated. Real use requires load cases, material selection, factor of safety, fastener checks, deflection review, and appropriate simulation or calculation.

10. Retrieval practice questions

These question-and-answer pairs are included to make the document useful for testing Atlas AI retrieval. Answers should be grounded in the sections above rather than guessed from general language patterns.

Question	Expected answer points
What is the recommended order for modelling a mechanical part?	Interpret function and manufacturing; select datums; create primary shape; add major cuts and interfaces; add secondary features; add finishing features; validate; create drawings and exports.
When should I use Extruded Boss/Base instead of Revolved Boss/Base?	Extrude is appropriate for prismatic geometry from a 2D profile; Revolve is appropriate for rotationally symmetric geometry around an axis.
What does a fully defined sketch mean?	The sketch has no remaining degrees of freedom; dimensions and relations fully constrain its geometry.
Why are fillets commonly added near the end?	They can alter edge topology and destabilise downstream references. Late placement can improve robustness, except where a fillet is functionally primary.
What is the difference between a clearance hole and a tapped hole?	A clearance hole allows a fastener shank to pass; a tapped hole contains internal threads to engage a screw.
Does STEP preserve the SOLIDWORKS feature tree?	Generally no. STEP is a neutral geometry exchange format and usually does not retain native parametric history.
What should I inspect first when a model fails after an edit?	Find the first failing feature in the FeatureManager tree and inspect its sketch and referenced geometry.
Why use datum planes instead of selecting arbitrary faces?	Datums can provide stable references tied to function and are less vulnerable to topology changes.

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